

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

3430U20-1



S19-3430U20-1

WEDNESDAY, 12 JUNE 2019 – MORNING**SCIENCE (Double Award)****Unit 2: CHEMISTRY 1
FOUNDATION TIER**

1 hour 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen.
Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	8	
3.	6	
4.	7	
5.	6	
6.	6	
7.	7	
8.	8	
9.	7	
Total	60	

3430U201
01

JUN193430U20101

BLANK PAGE

**PLEASE DO NOT WRITE
ON THIS PAGE**



1. Elements **A**, **B**, **C**, **D** and **E** are shown in the Periodic Table.

These letters are **not** the chemical symbols for the elements.

[illegible]

Give the **letter** of the element which fits each description below.

[5]

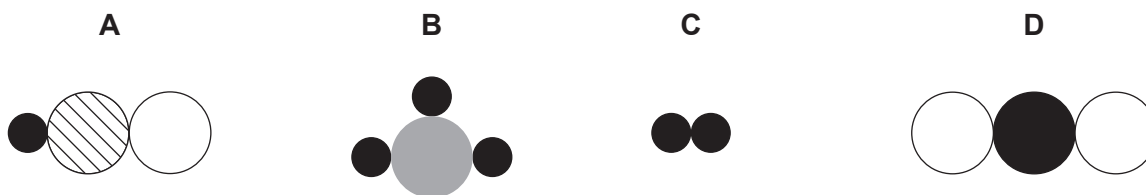
	Letter
is in Group 2	
makes a 'pop' noise with a lit splint	
has both metal and non-metal properties	
reacts with lithium to make lithium chloride	
is unreactive	

03
3430U201

5



2. Diagrams **A**, **B**, **C** and **D** represent hydrogen (H_2), sulfur dioxide (SO_2), hypochlorous acid (HClO) and phosphine (PH_3) but **not in that order**.



- (a) (i) Give the **letter** of the diagram that represents an element. Give a reason for your answer. [2]

Letter

Reason

- (ii) Give the **letter** of the diagram that represents SO_2 . [1]

.....

- (iii) Use diagrams **A**, **B**, **C** and **D** above to work out which of the following diagrams represents water (H_2O). **Circle** the correct diagram. [1]



- (b) (i) Calculate the relative molecular mass (M_r) of hypochlorous acid, HClO. [1]

$$A_r(\text{H}) = 1$$

$$A_r(\text{O}) = 16$$

$$A_r(\text{Cl}) = 35.5$$

$$M_r = \dots\dots\dots$$

- (ii) The relative molecular mass (M_r) of sulfur dioxide is 64.

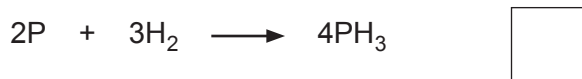
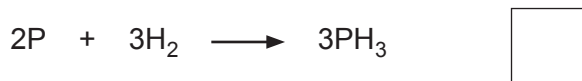
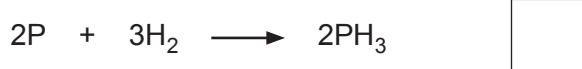
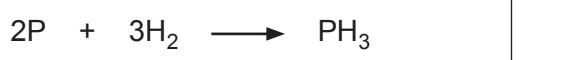
Calculate the percentage by mass of sulfur in sulfur dioxide, SO_2 . [2]

$$A_r(\text{S}) = 32$$

$$\text{Percentage} = \dots\dots\dots \%$$

- (c) Phosphine (PH_3) is made when phosphorus and hydrogen react.

Put a tick (✓) in the box next to the correctly balanced symbol equation for this reaction. [1]

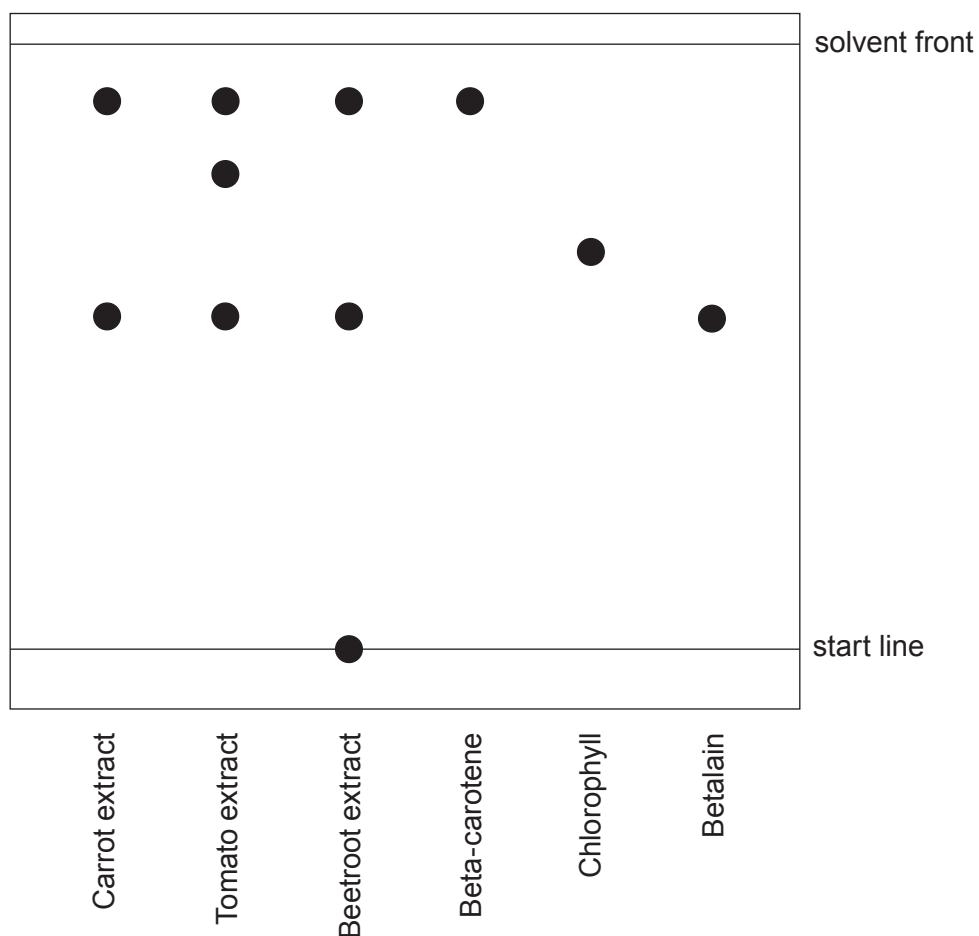


3. A student investigates the pigments found in some fruit and vegetables.

She obtains some coloured extracts from carrots, tomatoes and beetroot.

She places a spot of each extract on chromatography paper, along with spots of the three pigments beta-carotene, chlorophyll and betalain.

The diagram shows the chromatogram at the end of the experiment.



- (a) Give the names of **all** the substances which are **not** mixtures. Give a reason for your choice. [1]

Substances

.....

Reason

.....



- (b) Put a tick (✓) in the boxes next to the **two** conclusions that can be drawn from the chromatogram. [2]

chlorophyll is not present in carrot, tomato or beetroot extracts

☐

beta-carotene is present in carrot extract but not present in tomato extract

☐

both beta-carotene and betalain are present in beetroot extract

☐

betalain is present in tomato extract but not present in carrot extract

☐

both carrot and beetroot extracts contain a pigment other than beta-carotene, chlorophyll and betalain

☐

- (c) One of the pigments present in the carrot extract has travelled 4.4 cm above the start line.

The solvent front has travelled 8.0 cm. Calculate the R_f value of the pigment using the following equation. [2]

$$R_f = \frac{\text{distance travelled by pigment}}{\text{distance travelled by solvent front}}$$

$R_f =$

- (d) Give the reason why there is a spot remaining on the start line in the chromatogram for beetroot. [1]

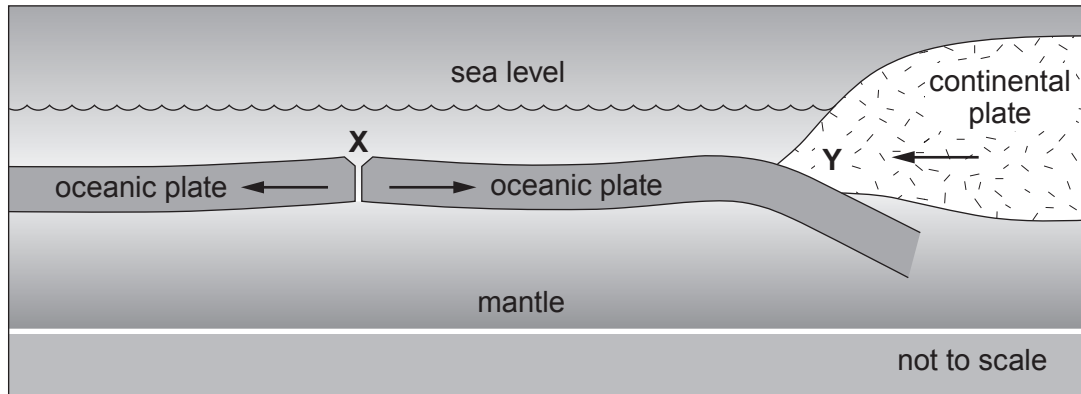
.....

.....



4. The Earth's crust is broken up into tectonic plates.

The diagram shows two plate boundaries **X** and **Y**.



(a) Describe what happens as the plates move apart at plate boundary **X**.

[2]

.....

.....

.....



- (b) The two types of plate have different densities as shown in the table.

Type of plate	Density (g/cm ³)
continental	4.7
oceanic	5.0

At boundary **Y** the oceanic plate is pushed **underneath** the continental plate.

- (i) State why the oceanic plate is pushed underneath the continental plate. [1]

.....

- (ii) State what happens to the oceanic plate when it goes underneath the continental plate. [1]

.....

- (iii) Underline the name of the type of plate boundary seen at **Y**. [1]

destructive

constructive

conservative

- (c) Using satellite imaging, scientists discovered that one oceanic plate had moved 537 cm over 600 years. Calculate the mean rate of movement of the plate over this time. [2]

$$\text{rate} = \frac{\text{distance}}{\text{time}}$$

Rate = cm/year

7



5. Planet **J** is similar in size to the Earth. However, the temperature on planet **J** is about 470°C and the clouds in its atmosphere are made of sulfuric acid.

A group of students investigated the properties of some metals. Their aim was to see if they could find a metal suitable for designing a spacecraft to explore planet **J**. Their findings are shown below.

Zinc – fizzes quite vigorously with sulfuric acid, has a melting point of 420°C and a density of 7.1 g/cm^3

Copper – has a melting point of 1083°C , does not react with sulfuric acid and has a density of 8.9 g/cm^3

Sodium – has a density of 1.0 g/cm^3 , reacts explosively with sulfuric acid and has a melting point of 98°C

Magnesium – has a density of 1.7 g/cm^3 , a melting point of 650°C and it fizzes vigorously with sulfuric acid

Titanium – does not react with sulfuric acid, has a melting point of 1675°C and a density of 4.5 g/cm^3

Lead – has a melting point of 328°C , a density of 11.3 g/cm^3 and does not react with sulfuric acid

The spacecraft needs to withstand the conditions on the surface of planet **J**. The mass of the spacecraft also needs to be as low as possible in order for it to have enough energy to escape the Earth's gravity.

- (a) Which **one** of these statements best describes why magnesium is an **unsuitable** metal for the spacecraft? Put a tick (✓) in the box next to the correct answer. [1]

its density is 1.7 g/cm^3

☐

its melting point is 650°C

☐

it fizzes vigorously with sulfuric acid

☐

it is malleable

☐

- (b) Small amounts of lead are sometimes used in electrical circuits.

Which **one** of these statements best describes why lead would **not** be suitable for use in the electrical circuits of the spacecraft? Put a tick (✓) in the box next to the correct answer. [1]

it does not react with sulfuric acid

☐

it is ductile

☐

it would melt when it lands on planet J

☐

its density is 11.3g/cm^3

☐

- (c) The students decided that titanium is the most suitable metal from which to build the spacecraft.

Put a tick (✓) in the boxes next to the **two** statements that best describe the reasons for their choice. [2]

it does not react with sulfuric acid

☐

it is expensive

☐

it is a good conductor of heat

☐

it is non-magnetic

☐

it has a melting point much higher than the temperature on planet J

☐

it is shiny so will reflect the sun's rays

☐

- (d) Sodium reacts explosively with sulfuric acid. Sodium sulfate and hydrogen are produced. Complete and balance the equation for this reaction. [2]



Examiner
only

6. Several areas of the UK add fluoride to drinking water.

State the benefit of fluoridation and discuss the reasons why some people are opposed to it.

[6 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

6

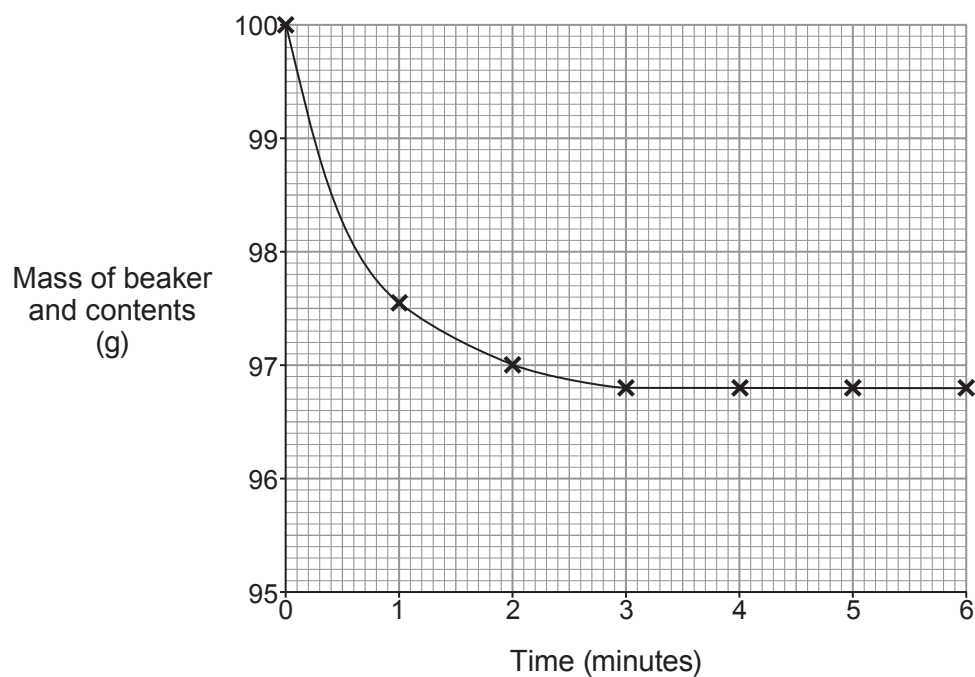


BLANK PAGE

**PLEASE DO NOT WRITE
ON THIS PAGE**



7. Powdered calcium carbonate was added to an excess of dilute hydrochloric acid in a beaker. The mass of the beaker and its contents was recorded every minute for 6 minutes. The graph shows the results.



The equation for the reaction is



- (a) Describe how the mass of the beaker and its contents changes over the first minute. Give the reason for this change. [2]

.....

.....



- (b) (i) Use the graph to determine the time taken for the reaction to finish. [1]

- (ii) When is the reaction at its fastest? Tick (✓) the correct box. [1]

from 0 – 0.5 minutes

☐

from 1 – 1.5 minutes

☐

from 2 – 2.5 minutes

☐

from 3 – 3.5 minutes

☐

- (c) Use your graph to calculate the mean rate of the reaction during the **first two minutes**.
Use the following equation. [2]

$$\text{rate} = \frac{\text{decrease in mass of beaker and contents (g)}}{\text{time (minutes)}}$$

Rate = g/minute

- (d) The experiment was repeated using the same mass of calcium carbonate but as a **lump** instead of a powder.

On the grid opposite, sketch the graph you would expect to obtain from this second experiment. [1]

7



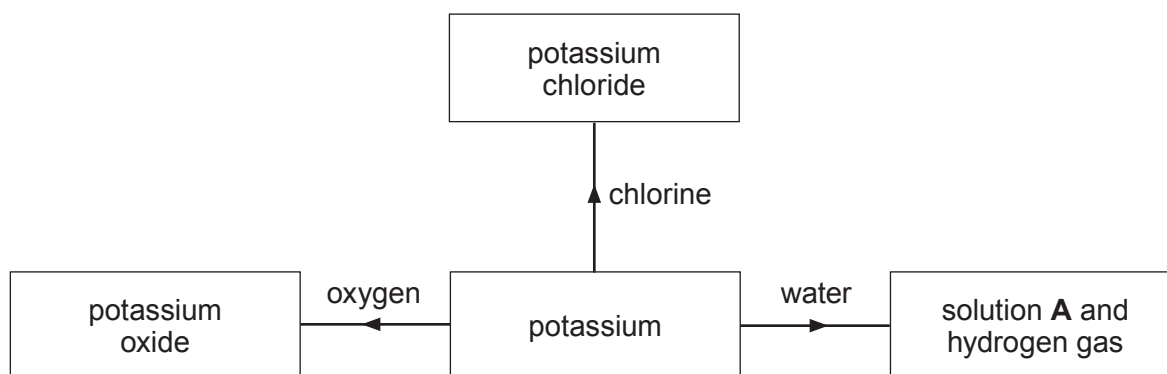
8. (a) The table gives information about some elements.

Element	Electronic structure	Group	Period
oxygen	2,6	6	2
chlorine		7	3
.....	2,8,5	5	3
potassium	2,8,8,1	1	

Complete the table.

[3]

- (b) The flow chart shows some of the reactions of potassium.



- (i) State **one** observation you would make when potassium reacts with water. [1]

.....

.....

- (ii) Apart from wearing gloves and safety goggles, give **one** safety precaution that should be taken when adding potassium to water. [1]

.....

.....



Examiner
only

(iii) Give the formula of solution **A**.

[1]

.....

(iv) Suggest a value for the pH of solution **A**.

[1]

.....

(v) Name a Group 1 metal that is **more** reactive than potassium.

[1]

.....

8



9. (a) The Earth's early atmosphere around 4 000 million years ago contained mainly carbon dioxide and water vapour produced by volcanoes.

- (i) Explain why the large percentage of water vapour in the Earth's atmosphere decreased over geological time. [2]

.....

.....

.....

- (ii) Give **two** reasons why the percentage of carbon dioxide in the Earth's atmosphere has decreased over geological time. [2]

.....

.....

.....

- (b) During the last 250 years the percentage of carbon dioxide in the Earth's atmosphere has increased from 0.03 % to 0.04 %. This has led to increased global warming. Give **one** reason for this increase and explain why global warming is a cause for concern. [2]

.....

.....

.....

.....

.....

.....

- (c) Ammonia present in the Earth's early atmosphere reacted with oxygen to produce nitrogen and water vapour. Complete the balancing of the symbol equation for this reaction. [1]



END OF PAPER

7



[illegible]

Examiner
only



BLANK PAGE

**PLEASE DO NOT WRITE
ON THIS PAGE**



BLANK PAGE

**PLEASE DO NOT WRITE
ON THIS PAGE**



BLANK PAGE

**PLEASE DO NOT WRITE
ON THIS PAGE**



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



24

1	H	
	Hydrogen	1

7	Li	9	Be
	Lithium		Beryllium
	3		4
23	Na	24	Mg
	Sodium		Magnesium
	11		12
39	K	40	Ca
	Potassium		Calcium
	19		20
86	Rb	88	Sr
	Rubidium		Strontium
	37		38
133	Cs	137	Ba
	Caesium		Barium
	55		56
223	Fr	226	Ra
	Francium		Radium
	87		88
		227	Ac
			Actinium
			89

11	B	12	C	14	N	16	O	19	F	20	Ne
	Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon
	5		6		7		8		9		10
27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar
	Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon
	13		14		15		16		17		18
70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
	31		32		33		34		35		36
115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
	Indium		Tin		Antimony		Tellurium		Iodine		Xenon
	49		50		51		52		53		54
204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
	Thallium		Lead		Bismuth		Polonium		Astatine		Radon
	81		82		83		84		85		86

65	Zn	63.5	Cu	59	Ni	59	Co	56	Fe	55	Mn	52	Cr	51	V	48	Ti	45	Sc
	Zinc		Copper		Nickel		Cobalt		Iron		Manganese		Chromium		Vanadium		Titanium		Scandium
	30		29		28		27		26		25		24		23		22		21
112	Cd	108	Ag	106	Pd	103	Rh	101	Ru	99	Tc	96	Mo	93	Nb	91	Zr	89	Y
	Cadmium		Silver		Palladium		Rhodium		Ruthenium		Technetium		Molybdenum		Niobium		Zirconium		Yttrium
	48		47		46		45		44		43		42		41		40		39
201	Hg	197	Au	195	Pt	192	Ir	190	Os	186	Re	184	W	181	Ta	179	Hf	139	La
	Mercury		Gold		Platinum		Iridium		Osmium		Rhenium		Tungsten		Tantalum		Hafnium		Lanthanum
	80		79		78		77		76		75		74		73		72		57
																			227
																			Ac
																			89

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number